

Title: Problem Sheet 2

Course: AERO40006 (Ordinary Differential Equations)

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Question 1: Find solutions of the following homogeneous second-order ODEs

$$(a) \frac{d^2y}{dx^2} + \frac{dy}{dx} + 4y = 0$$

$$y(0) = 2$$

$$\frac{dy}{dx}(0) = 2$$

$$\alpha^2 + \alpha + 4 = 0$$

$$\therefore b^2 - 4ac = -15 < 0$$

$$\therefore \lambda_1 = -\frac{b}{2a} = -1 - \frac{1}{2}$$

$$\lambda_2 = \frac{\sqrt{4ac - b^2}}{2a} = \frac{\sqrt{15}}{2}$$

$$\therefore y = e^{-\frac{1}{2}x} \left[A \cos\left(\frac{\sqrt{15}}{2}x\right) + B \sin\left(\frac{\sqrt{15}}{2}x\right) \right]$$

$$\therefore \frac{dy}{dx} = e^{-\frac{1}{2}x} \left[A \cos\left(\frac{\sqrt{15}}{2}x\right) + B \sin\left(\frac{\sqrt{15}}{2}x\right) \right]$$

$$+ e^{-\frac{1}{2}x} \left[-\frac{\sqrt{15}}{2} A \sin\left(\frac{\sqrt{15}}{2}x\right) + \frac{\sqrt{15}}{2} B \cos\left(\frac{\sqrt{15}}{2}x\right) \right]$$

$$= e^{-\frac{1}{2}x} \left[\left(\frac{\sqrt{15}}{2} B - A \right) \cos\left(\frac{\sqrt{15}}{2}x\right) + \left(-\frac{\sqrt{15}}{2} A + B \right) \sin\left(\frac{\sqrt{15}}{2}x\right) \right]$$

$$\therefore y(0) = 2 \quad \therefore A = 2$$

$$\therefore \frac{dy}{dx}(0) = 2 \quad \therefore \frac{\sqrt{15}}{2} B - A = 2 \quad \therefore B = \frac{8}{5\sqrt{15}}$$

$$\therefore y = e^{-\frac{1}{2}x} \left[2 \cos\left(\frac{\sqrt{15}}{2}x\right) + \frac{20}{15} \sin\left(\frac{\sqrt{15}}{2}x\right) \right]$$

$$(b) \frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 3y = 0$$

$$y(0) = 2$$

$$\frac{dy}{dx}(0) = 2$$

$$\alpha^2 + 2\alpha - 3 = 0$$

$$\therefore b^2 - 4ac = 16 > 0$$

$$\therefore \alpha^2 + 2\alpha - 3 = (\alpha - 1)(\alpha + 3) = 0 \quad \therefore \alpha_1 = 1, \alpha_2 = -3$$

$$\therefore y = Ae^x + Be^{-3x}$$

$$\therefore \frac{dy}{dx} = Ae^x - 3Be^{-3x}$$

$$\therefore y(0) = 2 \quad \therefore A + B = 2$$

$$\therefore \frac{dy}{dx}(0) = 2 \quad \therefore A - 3B = 2$$

$$\therefore A = 2$$

$$B = 0$$

$$\therefore y = 2e^x$$

$$(c) \frac{d^2y}{dx^2} - 12\frac{dy}{dx} + 36y = 0$$

$$y(0) = 1$$

$$\frac{dy}{dx}(0) = 1$$

$$\alpha^2 - 12\alpha + 36 = 0$$

$$\therefore b^2 - 4ac = 0$$

$$\therefore \alpha^2 - 12\alpha + 36 = (\alpha - 6)^2 = 0 \quad \therefore \alpha_1 = \alpha_2 = 6$$

$$\therefore y = (A + Bx)e^{6x}$$

$$\therefore \frac{dy}{dx} = Be^{6x} + 6(A + Bx)e^{6x} = (6A + B + 6Bx)e^{6x}$$

$$\therefore y(0) = 1 \quad \therefore A = 1$$

$$\therefore \frac{dy}{dx}(0) = 1 \quad \therefore 6A + B = 1 \quad \therefore B = -5$$

$$\therefore y = (1 - 5x)e^{6x}$$

$$(d) \frac{d^2y}{dx^2} + 10y = 0$$

$$y(0) = 0$$

$$\frac{dy}{dx}(0) = 2$$

$$\alpha^2 + 10 = 0$$

$$\therefore b^2 - 4ac = -40 < 0$$

$$\therefore \lambda_1 = -\frac{b}{2a} = 0$$

$$\lambda_2 = \frac{\sqrt{4ac - b^2}}{2a} = \sqrt{10}$$

$$\therefore y = e^{\alpha x} [A \cos(\sqrt{10}x) + B \sin(\sqrt{10}x)]$$

$$= A \cos(\sqrt{10}x) + B \sin(\sqrt{10}x)$$

$$\therefore \frac{dy}{dx} = -\sqrt{10}A \sin(\sqrt{10}x) + \sqrt{10}B \cos(\sqrt{10}x)$$

$$\therefore y(0) = 0 \quad \therefore A = 0$$

$$\therefore \frac{dy}{dx}(0) = 2 \quad \therefore \sqrt{10}B = 2 \quad \therefore B = \frac{2}{\sqrt{10}} = \frac{\sqrt{10}}{5}$$

$$\therefore y = \frac{\sqrt{10}}{5} \sin(\sqrt{10}x)$$

$$(e) \frac{d^2y}{dx^2} + \frac{dy}{dx} + y = 0$$

$$y(0) = 1$$

$$\frac{dy}{dx}(0) = 1$$

$$\alpha^2 + \alpha + 1 = 0$$

$$\therefore b^2 - 4ac = -3 < 0$$

$$\therefore \lambda_1 = -\frac{b}{2a} = -\frac{1}{2}, \lambda_2 = \frac{\sqrt{4ac - b^2}}{2a} = \frac{\sqrt{3}}{2}$$

$$\therefore y = e^{-\frac{1}{2}x} [A \cos(\frac{\sqrt{3}}{2}x) + B \sin(\frac{\sqrt{3}}{2}x)]$$

$$\therefore \frac{dy}{dx} = -\frac{1}{2}e^{-\frac{1}{2}x} [A \cos(\frac{\sqrt{3}}{2}x) + B \sin(\frac{\sqrt{3}}{2}x)]$$

$$+ e^{-\frac{1}{2}x} [-\frac{\sqrt{3}}{2}A \sin(\frac{\sqrt{3}}{2}x) + \frac{\sqrt{3}}{2}B \cos(\frac{\sqrt{3}}{2}x)]$$

$$= e^{-\frac{1}{2}x} [(-\frac{\sqrt{3}}{2}A - \frac{1}{2}B) \sin(\frac{\sqrt{3}}{2}x) + (\frac{\sqrt{3}}{2}B - \frac{1}{2}A) \cos(\frac{\sqrt{3}}{2}x)]$$

$$\therefore y(0) = 1 \quad \therefore A = 1$$

$$\therefore \frac{dy}{dx}(0) = 1 \quad \therefore \frac{\sqrt{3}}{2}B - \frac{1}{2}A = 1 \quad \therefore B = \sqrt{3}$$

$$\therefore y = e^{-\frac{1}{2}x} [\cos(\frac{\sqrt{3}}{2}x) + \sqrt{3} \sin(\frac{\sqrt{3}}{2}x)]$$

$$(f) \quad x^2 \frac{d^2 y}{dx^2} + 5x \frac{dy}{dx} + y = 0$$

$$y(1) = 0$$

$$\frac{dy}{dx}(1) = 3$$

$$\therefore (a-1)^2 - 4b = 12 > 0$$

$$\therefore \lambda = \frac{(1-a) \pm \sqrt{(a-1)^2 - 4b}}{2} = \frac{-4 \pm 2\sqrt{3}}{2} = -2 \pm \sqrt{3}$$

$$\therefore y = Ax^{-2+\sqrt{3}} + Bx^{-2-\sqrt{3}}$$

$$\therefore \frac{dy}{dx} = (-2+\sqrt{3})Ax^{-3+\sqrt{3}} + (-2-\sqrt{3})Bx^{-3-\sqrt{3}}$$

$$\therefore y(1) = 0 \quad \therefore A + B = 0$$

$$\therefore \frac{dy}{dx}(1) = 3 \quad \therefore (-2+\sqrt{3})A + (-2-\sqrt{3})B = 3$$

$$\therefore A = \frac{\sqrt{3}}{2}$$

$$B = -\frac{\sqrt{3}}{2}$$

$$\therefore y = \frac{\sqrt{3}}{2}x^{-2+\sqrt{3}} - \frac{\sqrt{3}}{2}x^{-2-\sqrt{3}}$$

$$(g) \quad x^2 \frac{d^2 y}{dx^2} + 8x \frac{dy}{dx} - 2y = 0$$

$$y(1) = 2$$

$$\frac{dy}{dx}(1) = 0$$

$$\therefore (a-1)^2 - 4b = \overset{49+8}{4-32} = \overset{57}{-23} < 0$$

$$\therefore \lambda_1 = \frac{1-a}{2} = \frac{3}{2}$$

$$\lambda_2 = \frac{\sqrt{4b - (a-1)^2}}{2} = \frac{\sqrt{23}}{2}$$

$$\therefore y = x^{\frac{3}{2}} \left[A \cos\left(\frac{\sqrt{23}}{2} \ln(x)\right) + B \sin\left(\frac{\sqrt{23}}{2} \ln(x)\right) \right]$$

$$\begin{aligned} \therefore \frac{dy}{dx} &= \frac{3}{2} x^{\frac{1}{2}} \left[A \cos\left(\frac{\sqrt{23}}{2} \ln(x)\right) + B \sin\left(\frac{\sqrt{23}}{2} \ln(x)\right) \right] \\ &\quad + x^{\frac{3}{2}} \left[-\frac{\sqrt{23}}{2x} A \sin\left(\frac{\sqrt{23}}{2} \ln(x)\right) + \frac{\sqrt{23}}{2x} B \cos\left(\frac{\sqrt{23}}{2} \ln(x)\right) \right] \\ &= x^{\frac{3}{2}} \left[\left(\frac{3}{2}B - \frac{\sqrt{23}}{2x} A\right) \sin\left(\frac{\sqrt{23}}{2} \ln(x)\right) + \left(\frac{3}{2}A + \frac{\sqrt{23}}{2x} B\right) \cos\left(\frac{\sqrt{23}}{2} \ln(x)\right) \right] \end{aligned}$$

$$\therefore y(1) = 2 \quad \therefore A = 2$$

$$\therefore \frac{dy}{dx}(1) = 0 \quad \therefore \frac{3}{2}A + \frac{\sqrt{23}}{2}B = 0 \quad \therefore B = -\frac{6}{23}\sqrt{23}$$

$$\therefore y = x^{\frac{3}{2}} \left[2 \cos\left(\frac{\sqrt{23}}{2} \ln(x)\right) - \frac{6}{23}\sqrt{23} \sin\left(\frac{\sqrt{23}}{2} \ln(x)\right) \right]$$

$$y = \left(1 + \frac{7\sqrt{57}}{57}\right) x^{\frac{-7+\sqrt{57}}{2}} + \left(1 - \frac{7\sqrt{57}}{57}\right) x^{\frac{-7-\sqrt{57}}{2}}$$

$$(h) \quad x^2 \frac{d^2 y}{dx^2} + 9x \frac{dy}{dx} + 16y = 0$$

$$y(1) = 0$$

$$\frac{dy}{dx}(1) = 2$$

$$\therefore (a-1)^2 - 4b = 0$$

$$\therefore \lambda_1 = \frac{a-1}{2} = 4$$

$$\therefore y = (A + B \ln(x)) x^4$$

$$\begin{aligned} \therefore \frac{dy}{dx} &= 4(A + B \ln(x)) x^3 + \left(\frac{B}{x}\right) x^4 \\ &= (4A + 4B \ln(x) + \frac{B}{x}) x^4 \end{aligned}$$

$$\therefore y(1) = 0 \quad \therefore A = 0$$

$$\therefore \frac{dy}{dx}(1) = 2 \quad \therefore 4A + B = 2 \quad \therefore B = 2$$

$$\therefore y = 2 \ln(x) x^4$$

Question 2: Find solutions of the following inhomogeneous second-order ODEs

$$(a) \frac{d^2y}{dx^2} + 3\frac{dy}{dx} - y = 7e^{2x} \cos(4x) \quad y(0) = 0 \quad \frac{dy}{dx}(0) = 0$$

$$y = y_h + y_p$$

$$\text{For } y_h, \alpha^2 + 3\alpha - 1 = 0$$

$$\therefore b^2 - 4ac = 13 > 0 \quad \therefore \alpha = \frac{-3 \pm \sqrt{13}}{2}$$

$$\therefore y_h = Ae^{\frac{-3+\sqrt{13}}{2}x} + Be^{\frac{-3-\sqrt{13}}{2}x}$$

$$\therefore \frac{dy_h}{dx} = \frac{-3+\sqrt{13}}{2} Ae^{\frac{-3+\sqrt{13}}{2}x} + \frac{-3-\sqrt{13}}{2} Be^{\frac{-3-\sqrt{13}}{2}x}$$

$$\text{For } y_p, \text{ Let } y_p = e^{2x} [A \sin(4x) + B \cos(4x)]$$

$$\therefore y_p' = 2e^{2x} [A \sin(4x) + B \cos(4x)] + e^{2x} [4A \cos(4x) - 4B \sin(4x)]$$

$$= e^{2x} [(2A - 4B) \sin(4x) + (2B + 4A) \cos(4x)]$$

$$\therefore y_p'' = 2e^{2x} [(2A - 4B) \sin(4x) + (2B + 4A) \cos(4x)]$$

$$+ e^{2x} [(8A - 16B) \cos(4x) - (8B + 16A) \sin(4x)]$$

$$= e^{2x} [(-12A - 16B) \sin(4x) + (16A - 12B) \cos(4x)]$$

Question 2: Find solutions of the following inhomogeneous second-order ODEs

$$(a) \quad \frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} - y = 7e^{2x} \cos(4x) \quad y(0) = 0 \quad \frac{dy}{dx}(0) = 0$$

$$e^{2x} [(-12A - 16B) \sin(4x) + (16A - 12B) \cos(4x)]$$

$$+ 3e^{2x} [(2A - 4B) \sin(4x) + (2B + 4A) \cos(4x)]$$

$$- e^{2x} [A \sin(4x) + B \cos(4x)] = 7e^{2x} \cos(4x)$$

$$\therefore e^{2x} [(-7A - 28B) \sin(4x) + (28A - 7B) \cos(4x)]$$

$$= 7e^{2x} [(-A - 4B) \sin(4x) + (4A - B) \cos(4x)] = 7e^{2x} \cos(4x)$$

$$\therefore (-A - 4B) \sin(4x) = 0$$

$$(4A - B) \cos(4x) = \cos(4x)$$

$$\therefore -A - 4B = 0$$

$$4A - B = 1$$

$$\therefore A = \frac{4}{17}, \quad B = -\frac{1}{17}$$

$$\therefore y_p = e^x \left[\frac{4}{17} \sin(4x) - \frac{1}{17} \cos(4x) \right]$$

$$\therefore \frac{dy_p}{dx} = \left[\left(\frac{4}{17} + \frac{4}{17} \right) \sin(4x) + \left(-\frac{1}{17} + \frac{16}{17} \right) \cos(4x) \right] e^x$$

Question 2: Find solutions of the following inhomogeneous second-order ODEs

$$(a) \frac{d^2y}{dx^2} + 3\frac{dy}{dx} - y = 7e^{2x} \cos(4x) \quad y(0) = 0 \quad \frac{dy}{dx}(0) = 0$$

$$\therefore y = Ae^{\frac{-3+\sqrt{13}}{2}x} + Be^{\frac{-3-\sqrt{13}}{2}x} + e^x \left[\frac{4}{17} \sin(4x) - \frac{1}{17} \cos(4x) \right]$$

$$\begin{aligned} \therefore \frac{dy}{dx} &= \frac{-3+\sqrt{13}}{2} Ae^{\frac{-3+\sqrt{13}}{2}x} + \frac{-3-\sqrt{13}}{2} Be^{\frac{-3-\sqrt{13}}{2}x} \\ &\quad + e^x \left[\frac{8}{17} \sin(4x) + \frac{15}{17} \cos(4x) \right] \end{aligned}$$

$$\therefore y(0) = 0 \quad \therefore A + B - \frac{1}{17} = 0$$

$$\therefore \frac{dy}{dx}(0) = 0 \quad \therefore \frac{-3+\sqrt{13}}{2} A + \frac{-3-\sqrt{13}}{2} B + \frac{15}{17} = 0$$

$$\therefore A = -0.1908$$

$$B = 0.2497$$

$$\begin{aligned} \therefore y &= -0.0578 e^{\frac{-3+\sqrt{13}}{2}x} - 0.8247 e^{\frac{-3-\sqrt{13}}{2}x} \\ &\quad + e^x \left[\frac{4}{17} \sin(4x) - \frac{1}{17} \cos(4x) \right] \end{aligned}$$

$$(b) \frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 8y = 3e^x$$

$$y(0) = 2$$

$$\frac{dy}{dx}(0) = 1$$

$$y = y_h + y_p$$

For y_h ,

$$\alpha^2 - 6\alpha + 8 = 0 \quad \therefore b^2 - 4ac = 4 > 0$$

$$\therefore \alpha = \frac{6 \pm 2}{2} = 4 \text{ OR } 2$$

$$\therefore y_h = Ae^{4x} + Be^{2x}$$

$$\therefore \frac{dy_h}{dx} = 4Ae^{4x} + 2Be^{2x}$$

For y_p ,

$$\text{Let } y_p = Ae^x$$

$$\therefore y_p' = Ae^x \quad \therefore y_p'' = Ae^x$$

$$\therefore 3Ae^x = 3e^x \quad \therefore A = 1$$

$$\therefore y_p = e^x \quad \therefore \frac{dy_p}{dx} = e^x$$

$$(b) \frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 8y = 3e^x$$

$$y(0) = 2$$

$$\frac{dy}{dx}(0) = 1$$

$$\therefore y(0) = A + B + 1 = 2$$

$$\therefore \frac{dy}{dx}(0) = 4A + 2B + 1 = 1$$

$$\therefore A = -\frac{1}{2} - 1$$

$$\therefore B = 1 - 2$$

$$\therefore y = -\frac{1}{2}e^{4x} + e^{2x} + e^x$$

$$(c) \frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 10\sin(x) \quad y(0) = 0 \quad \frac{dy}{dx}(0) = 0$$

$$y = y_h + y_p$$

For y_h ,

$$\alpha^2 - 3\alpha + 2 = 0 \quad \therefore b^2 - 4ac = 1 > 0$$

$$\therefore (\alpha - 1)(\alpha - 2) = 0 \quad \therefore \alpha = 1 \text{ OR } 2$$

$$\therefore y_h = Ae^x + Be^{2x}$$

$$\therefore y_h' = Ae^x + 2Be^{2x}$$

For y_p ,

$$\text{Let } y_p = A\sin(x) + B\cos(x)$$

$$\therefore y_p' = A\cos(x) - B\sin(x)$$

$$\therefore y_p'' = -A\sin(x) - B\cos(x)$$

$$(c) \frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 10\sin(x) \quad y(0) = 0 \quad \frac{dy}{dx}(0) = 0$$

$$\therefore (-A + 3B + 2A)\sin(x) + (-B - 3A + 2B)\cos(x)$$

$$= (3B + A)\sin(x) + (B - 3A)\cos(x) = 10\sin(x)$$

$$\therefore 3B + A = 10$$

$$B - 3A = 0$$

$$\therefore B = 3$$

$$A = 1$$

$$\therefore y_p = \sin(x) + 3\cos(x) \quad \therefore \frac{dy_p}{dx} = -3\sin(x) + \cos(x)$$

$$\therefore y(0) = A + B + 3 = 0$$

$$\frac{dy}{dx}(0) = A + 2B + 1 = 0$$

$$\therefore A = -5$$

$$B = 2$$

$$\therefore y = -5e^x + 2e^{2x} + \sin(x) + 3\cos(x)$$

$$(d) \frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 2x^2 + 5$$

$$y(0) = 0$$

$$\frac{dy}{dx}(0) = 0$$

$$y = y_h + y_p$$

For y_h ,

$$\alpha^2 - \alpha - 2 = 0 \quad \therefore b^2 - 4ac = 9 > 0$$

$$\therefore (\alpha - 2)(\alpha + 1) = 0 \quad \therefore \alpha = 2 \text{ OR } -1$$

$$\therefore y_h = Ae^{2x} + Be^{-x}$$

$$\therefore y_h' = 2Ae^{2x} - Be^{-x}$$

For y_p ,

$$\text{Let } y_p = Ax^2 + Bx + C$$

$$\therefore y_p' = 2Ax + B$$

$$\therefore y_p'' = 2A$$

$$(d) \frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 2x^2 + 5$$

$$y(0) = 0$$

$$\frac{dy}{dx}(0) = 0$$

$$\therefore 2A - 2Ax - B - 2Ax^2 - 2Bx - 2C = 2x^2 + 5$$

$$\therefore -2A = 2$$

$$-2B - 2A = 0$$

$$2A - B - 2C = 5$$

$$\therefore A = -1$$

$$B = 1$$

$$C = -4$$

$$\therefore y_p = -x^2 + x - 4 \quad \therefore \frac{dy_p}{dx} = -2x + 1$$

$$\therefore y(0) = A + B - 4 = 0$$

$$\therefore \frac{dy}{dx}(0) = 2A - B + 1 = 0$$

$$\therefore A = 1, B = 3$$

$$\therefore y = e^{2x} + 3e^{-x} - x^2 + x - 4$$

$$(e) \frac{d^2y}{dx^2} - 4y = 5 \sinh(2x)$$

$$y(0) = 1$$

$$\frac{dy}{dx}(0) = 0$$

$$y = y_h + y_p$$

For y_h ,

$$a^2 - 4 = 0 \quad \therefore b^2 - 4ac = 16 > 0$$

$$\therefore \alpha = \pm 2$$

$$\therefore y_h = Ae^{2x} + Be^{-2x}$$

$$\therefore y_h' = 2Ae^{2x} - 2Be^{-2x}$$

For y_p , let $y_p = xA \sinh(2x) + xB \cosh(2x)$

$$\begin{aligned} \therefore y_p' &= 2xA \cosh(2x) + A \sinh(2x) + 2xB \sinh(2x) + B \cosh(2x) \\ &= (A + 2xB) \sinh(2x) + (B + 2xA) \cosh(2x) \end{aligned}$$

$$\begin{aligned} \therefore y_p'' &= 2B \sinh(2x) + (2A + 4xB) \cosh(2x) \\ &\quad + 2A \cosh(2x) + (2B + 4xA) \sinh(2x) \\ &= (4B + 4xA) \sinh(2x) + (4A + 4xB) \cosh(2x) \end{aligned}$$

$$(e) \frac{d^2 y}{dx^2} - 4y = 5 \sinh(2x)$$

$$y(0) = 1$$

$$\frac{dy}{dx}(0) = 0$$

$$\therefore 4B \sinh(2x) + 4A \cosh(2x) = 5 \sinh(2x)$$

$$\therefore A = 0$$

$$B = \frac{5}{4}$$

$$\therefore y_p = \frac{5}{4} x \cosh(2x)$$

$$\therefore \frac{dy_p}{dx} = \frac{5}{4} \cosh(2x) + \frac{5}{2} x \sinh(2x)$$

$$\therefore y(0) = A + B = 1$$

$$\frac{dy}{dx}(0) = 2A - 2B + \frac{5}{4} = 0$$

$$\therefore A = -\frac{3}{16} \quad \frac{3}{16}$$

$$B = \frac{13}{16} \quad \frac{13}{16}$$

$$\therefore y = -\frac{3}{16} e^{2x} + \frac{13}{16} e^{-2x} + \frac{5}{4} x \cosh(2x)$$

$$(f) \frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 5y = 21e^{2x}$$

$$y(0) = 0$$

$$\frac{dy}{dx}(0) = 1$$

$$y = y_h + y_p$$

For y_h ,

$$a^2 - 4a + 5 = 0 \quad \because b^2 - 4ac = -4 < 0$$

$$\therefore \lambda_1 = -\frac{b}{2a} = 2$$

$$\lambda_2 = \frac{\sqrt{4ac - b^2}}{2a} = 1$$

$$\therefore y_h = e^{2x} [A \cos(x) + B \sin(x)]$$

$$\therefore y_h' = 2e^{2x} [A \cos(x) + B \sin(x)] + e^{2x} [-A \sin(x) + B \cos(x)]$$

$$= e^{2x} [(2A + B) \cos(x) + (2B - A) \sin(x)]$$

For y_p ,

$$\text{Let } y_p = Ae^{2x}$$

$$\therefore y_p' = 2Ae^{2x}$$

$$\therefore y_p'' = 4Ae^{2x}$$

$$(f) \quad \frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 5y = 21e^{2x} \qquad y(0) = 0 \qquad \frac{dy}{dx}(0) = 1$$

$$\therefore 4Ae^{2x} - 8Ae^{2x} + 5Ae^{2x} = Ae^{2x} = 21e^{2x}$$

$$\therefore A = 21$$

$$\therefore y_p = 21e^{2x} \qquad \therefore y_p' = 42e^{2x}$$

$$\therefore y(0) = A + 21 = 0 \qquad \therefore A = -21$$

$$\frac{dy}{dx}(0) = 2A + B + 42 = 1 \qquad \therefore B = 1$$

$$\therefore y = e^x [-21 \cos(x) + \sin(x)] + 21e^{2x}$$

Question 3: Derive Laplace Transforms of the following functions

(a) $f(t) = t^3$

(c) $f(t) = t^2 e^t$

(b) $f(t) = \cosh(at)$

(d) $f(t) = t \cos(bt)$

(a) $L\{f(t)\} = \int_0^\infty t^3 e^{-st} dt$

$$= \left[-\frac{1}{s} t^3 e^{-st} - \frac{3}{s^2} t^2 e^{-st} - \frac{6}{s^3} t e^{-st} - \frac{6}{s^4} e^{-st} \right]_0^\infty$$

$$= \left[-\frac{e^{-st}}{s} \left(1 + \frac{3}{s} + \frac{6}{s^2} + \frac{6}{s^3} \right) \right]_0^\infty$$

$$= \frac{1}{s} + \frac{3}{s^2} + \frac{6}{s^3} + \frac{6}{s^4}$$

D	I
+	$t^3 e^{-st}$
-	$3t^2 \cdot \frac{1}{s} e^{-st}$
+	$6t \cdot \frac{1}{s^2} e^{-st}$
-	$6 \cdot \frac{1}{s^3} e^{-st}$
+	$0 \cdot \frac{1}{s^4} e^{-st}$

(b) $L\{f(t)\} = \int_0^\infty \cosh(at) e^{-st} dt$

$$\therefore \int \cosh(at) e^{-st} dt = -\frac{1}{s} \cosh(at) e^{-st} + \frac{a}{s^2} \sinh(at) e^{-st} - \frac{a}{s^3} \int \cosh(at) dt$$

D	I
+	$\cosh(at) e^{-st}$
-	$a \sinh(at) \cdot \frac{1}{s} e^{-st}$
+	$a^2 \cosh(at) \cdot \frac{1}{s^2} e^{-st}$
-	$a^3 \sinh(at) \cdot \frac{1}{s^3} e^{-st}$
...	...

$$\therefore \left(1 + \frac{a^2}{s^2} \right) \int \cosh(at) dt = -\frac{1}{s} \cosh(at) e^{-st} + \frac{a}{s^2} \sinh(at) e^{-st}$$

$$\therefore L\{f(t)\} = \left[-\frac{s}{s^2+a^2} \cosh(at) e^{-st} + \frac{a}{s^2+a^2} \sinh(at) e^{-st} \right]_0^\infty$$

$$= \frac{s}{s^2+a^2}$$

Question 3: Derive Laplace Transforms of the following functions

(a) $f(t) = t^3$

(c) $f(t) = t^2 e^t$

(b) $f(t) = \cosh(at)$

(d) $f(t) = t \cos(bt)$

$$(a) \quad L\{f'\} = sL\{f\} - f(0)$$

$$\therefore L\{3t^2\} = 3L\{t^2\} = sL\{t^3\} - 0$$

$$\therefore \frac{6}{s^3} = sL\{t^3\}$$

$$\therefore L\{t^3\} = \frac{6}{s^4}$$

$$(b) \quad L\{f''\} = s^2 L\{f\} - sf(0) - f'(0)$$

.....

$$(d) \quad \dots\dots$$

Question 3: Derive Laplace Transforms of the following functions

(a) $f(t) = t^3$

(c) $f(t) = t^2 e^t$

(b) $f(t) = \cosh(at)$

(d) $f(t) = t \cos(bt)$

$$\begin{aligned}
 \text{(c)} \quad \mathcal{L}\{f(t)\} &= \int_0^{\infty} t e^t e^{-st} dt & \mathcal{D} & & \mathcal{I} \\
 &= \int_0^{\infty} t e^{(1-s)t} dt & + t^2 & & e^{(1-s)t} \\
 &= \left[\frac{t^2}{1-s} e^{(1-s)t} - \frac{2t}{(1-s)^2} e^{(1-s)t} \right. & - 2t & & \frac{1}{1-s} e^{(1-s)t} \\
 &\quad \left. + \frac{2}{(1-s)^3} e^{(1-s)t} \right]_0^{\infty} & + 2 & & \frac{1}{(1-s)^2} e^{(1-s)t} \\
 &= \left[\frac{e^{(1-s)t}}{1-s} \left(t^2 - \frac{2t}{1-s} + \frac{2}{(1-s)^2} \right) \right]_0^{\infty} & - 0 & & \frac{1}{(1-s)^3} e^{(1-s)t} \\
 &= \frac{-2}{(1-s)^3}
 \end{aligned}$$

Question 3: Derive Laplace Transforms of the following functions

(a) $f(t) = t^3$

(c) $f(t) = t^2 e^t$

(b) $f(t) = \cosh(at)$

(d) $f(t) = t \cos(bt)$

(d) $L\{f(t)\} = \int_0^{\infty} t \cos(bt) e^{-st} dt$

$\therefore \int t \cos(bt) e^{-st} dt$

$= -\frac{t}{s} e^{-st} + \frac{1}{s} \cos(bt) e^{-st} - \frac{bt}{s} \sin(bt) e^{-st}$
 $+ \frac{b^2}{s^2} \int t \cos(bt) e^{-st} dt$

$\therefore L\{f(t)\} = \frac{e^{-st}}{s^2 - b^2} [-st + \cos(bt) - bt \sin(bt)]_0^{\infty}$

$= \frac{1}{s^2 - b^2} (-1)$

$= \frac{1}{b^2 - s^2}$

	D	I
	$+ t \cos(bt)$	e^{-st}
	$- \cos(bt) - bt \sin(bt)$	$-\frac{1}{s} e^{-st}$
	$+ b \sin(bt) - b \sin(bt)$	$-\frac{1}{s} e^{-st}$
	$- b^2 t \cos(bt)$	
	...	...

Question 4: Find inverse Laplace Transforms of the following functions

$$(a) F(s) = \frac{1}{s^2 - 4s + 3}$$

$$(c) F(s) = \frac{s-3}{(s-3)^2 + 9}$$

$$(b) F(s) = \frac{2s}{s^2 - 48}$$

$$(d) F(s) = \frac{4}{s^2 + 9} - \frac{1}{(s-3)^2}$$

$$(a) f(t) = L^{-1}\{F(s)\} = L^{-1}\left\{\frac{1}{s^2 - 4s + 3}\right\}$$

$$= L^{-1}\left\{\frac{-\frac{1}{2}}{s-1}\right\} + L^{-1}\left\{\frac{\frac{1}{2}}{s-3}\right\}$$

$$= -\frac{1}{2}L^{-1}\left\{\frac{1}{s-1}\right\} + \frac{1}{2}L^{-1}\left\{\frac{1}{s-3}\right\}$$

$$= -\frac{1}{2}e^t + \frac{1}{2}e^{3t}$$

$$(b) f(t) = L^{-1}\left\{\frac{2s}{s^2 - 48}\right\} = L^{-1}\left\{\frac{1}{s - 4\sqrt{3}} + \frac{1}{s + 4\sqrt{3}}\right\}$$

$$= L^{-1}\left\{\frac{1}{s - 4\sqrt{3}}\right\} + L^{-1}\left\{\frac{1}{s + 4\sqrt{3}}\right\}$$

$$= e^{4\sqrt{3}t} + e^{-4\sqrt{3}t}$$

Question 4: Find inverse Laplace Transforms of the following functions

(a) $F(s) = \frac{1}{s^2 - 4s + 3}$

(c) $F(s) = \frac{s-3}{(s-3)^2 + 9}$

(b) $F(s) = \frac{2s}{s^2 - 48}$

(d) $F(s) = \frac{4}{s^2 + 9} - \frac{1}{(s-3)^2}$

(c) $f(t) = L^{-1}\{F(s)\} = L^{-1}\left\{\frac{s-3}{(s-3)^2 + 9}\right\}$

Remove the shift

$$= L^{-1}\left\{\frac{s-3}{s^2 - 6s + 18}\right\} = L^{-1}\left\{\frac{\frac{1}{2}}{s-3-3i} + \frac{\frac{1}{2}}{s-3+3i}\right\}$$

$f^*(t)$

$= L^{-1}\left\{\frac{s}{s^2 + 9}\right\}$

$= \cos(3t)$

$\therefore f(t) = e^{3t} \cos(3t)$

$$= \frac{1}{2} L^{-1}\left\{\frac{1}{s-3-3i}\right\} + \frac{1}{2} L^{-1}\left\{\frac{1}{s-3+3i}\right\}$$

$$= \frac{1}{2} e^{(3+3i)t} + \frac{1}{2} e^{(3-3i)t}$$

(d) $f(t) = L^{-1}\{F(s)\} = L^{-1}\left\{\frac{4}{s^2 + 9} - \frac{1}{(s-3)^2}\right\}$

$$= 4 L^{-1}\left\{\frac{1}{s^2 + 9}\right\} - L^{-1}\left\{\frac{1}{(s-3)^2}\right\}$$

$$= 4e^{-9t} - te^{3t}$$

$$\frac{4}{3} \sin(3t) - te^{3t}$$

Question 5: Solve the following ODEs using Laplace Transforms

$$\begin{array}{lll} \text{(a)} \quad \frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} - 4y = e^{-x} & y(0) = 0 & \frac{dy}{dx}(0) = 0 \\ \text{(b)} \quad \frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} - 3y = e^{-3x} & y(0) = 0 & \frac{dy}{dx}(0) = 0 \\ \text{(c)} \quad \frac{d^2 y}{dx^2} - \frac{dy}{dx} - 6y = \cos(2x) & y(0) = 0 & \frac{dy}{dx}(0) = 0 \end{array}$$

$$\text{(a)} \quad L\left\{\frac{dy}{dx}\right\} = sL\{y(x)\} - y(0) = sL\{y(x)\}$$

$$L\left\{\frac{d^2 y}{dx^2}\right\} = sL\left\{\frac{dy}{dx}\right\} - \frac{dy}{dx}(0)$$

$$= s^2 L\{y(x)\} - sy(0) - \frac{dy}{dx}(0) = s^2 L\{y(x)\}$$

$$\therefore s^2 L\{y(x)\} + 3sL\{y(x)\} - 4L\{y(x)\} = L\{e^{-x}\}$$

$$\therefore (s^2 + 3s - 4)L\{y(x)\} = L\{e^{-x}\} = \frac{1}{s+1}$$

$$\therefore L\{y(x)\} = \frac{1}{(s+4)(s-1)(s+1)} = \frac{\frac{1}{15}}{s+4} + \frac{\frac{1}{10}}{s-1} + \frac{-\frac{1}{6}}{s+1}$$

$$\therefore y(x) = \frac{1}{15} L^{-1}\left\{\frac{1}{s+4}\right\} + \frac{1}{10} L^{-1}\left\{\frac{1}{s-1}\right\} - \frac{1}{6} L^{-1}\left\{\frac{1}{s+1}\right\}$$

$$= \frac{1}{15} e^{-4t} + \frac{1}{10} e^t - \frac{1}{6} e^{-t}$$

Question 5: Solve the following ODEs using Laplace Transforms

- (a) $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} - 4y = e^{-x}$ $y(0) = 0$ $\frac{dy}{dx}(0) = 0$
- (b) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 3y = e^{-3x}$ $y(0) = 0$ $\frac{dy}{dx}(0) = 0$
- (c) $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 6y = \cos(2x)$ $y(0) = 0$ $\frac{dy}{dx}(0) = 0$

$$\text{b) } L\left\{\frac{dy}{dx}\right\} = sL\{y\} - y(0) = sL\{y\}$$

$$\begin{aligned} L\left\{\frac{d^2y}{dx^2}\right\} &= sL\left\{\frac{dy}{dx}\right\} - \frac{dy}{dx}(0) = s^2L\{y\} - sy(0) - \frac{dy}{dx}(0) \\ &= s^2L\{y\} \end{aligned}$$

$$\therefore s^2L\{y\} + 2sL\{y\} - 3L\{y\} = L\{e^{-3x}\}$$

$$\therefore (s^2 + 2s - 3)L\{y\} = \frac{1}{s+3}$$

$$\begin{aligned} \therefore L\{y\} &= \frac{1}{(s-1)(s+3)(s+3)} = \frac{\frac{1}{4}}{(s-1)(s+3)} + \frac{-\frac{1}{4}}{(s+3)^2} \\ &= \frac{\frac{1}{16}}{s-1} + \frac{-\frac{1}{16}}{s+3} + \frac{-\frac{1}{4}}{(s+3)^2} \end{aligned}$$

$$\begin{aligned} \therefore y &= \frac{1}{16} L^{-1}\left\{\frac{1}{s-1}\right\} - \frac{1}{16} L^{-1}\left\{\frac{1}{s+3}\right\} - \frac{1}{4} L^{-1}\left\{\frac{1}{(s+3)^2}\right\} \\ &= \frac{1}{16} e^t - \frac{1}{16} e^{-3t} - \frac{1}{4} t e^{-3t} \end{aligned}$$

Question 5: Solve the following ODEs using Laplace Transforms

$$\begin{array}{lll} \text{(a)} \quad \frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} - 4y = e^{-x} & y(0) = 0 & \frac{dy}{dx}(0) = 0 \\ \text{(b)} \quad \frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} - 3y = e^{-3x} & y(0) = 0 & \frac{dy}{dx}(0) = 0 \\ \text{(c)} \quad \frac{d^2 y}{dx^2} - \frac{dy}{dx} - 6y = \cos(2x) & y(0) = 0 & \frac{dy}{dx}(0) = 0 \end{array}$$

$$\text{(c)} \quad L\left\{\frac{dy}{dx}\right\} = sL\{y\} - y(0) = sL\{y\}$$

$$L\left\{\frac{d^2 y}{dx^2}\right\} = sL\left\{\frac{dy}{dx}\right\} - \frac{dy}{dx}(0)$$

$$= s^2 L\{y\} - sy(0) - \frac{dy}{dx}(0) = s^2 L\{y\}$$

$$\therefore s^2 L\{y\} - sL\{y\} - 6L\{y\} = L\{\cos(2x)\}$$

$$\therefore (s^2 - s - 6)L\{y\} = \frac{s}{s^2 + 4}$$

$$\begin{aligned} \therefore L\{y\} &= \frac{s}{(s+2)(s-3)(s^2+4)} = \frac{A}{s+2} + \frac{B}{s-3} + \frac{Cs+D}{s^2+4} \\ &= \frac{\frac{1}{20}}{s+2} + \frac{\frac{3}{65}}{s-3} + \frac{-\frac{5}{52}s - \frac{1}{26}}{s^2+4} \end{aligned}$$

$$\therefore y = \frac{1}{20} L^{-1}\left\{\frac{1}{s+2}\right\} + \frac{3}{65} L^{-1}\left\{\frac{1}{s-3}\right\} - \frac{5}{52} L^{-1}\left\{\frac{s}{s^2+4}\right\} - \frac{1}{52} L^{-1}\left\{\frac{2}{s^2+4}\right\}$$

$$= \frac{1}{20} e^{-2x} + \frac{3}{65} e^{3x} - \frac{5}{52} \cos(2x) - \frac{1}{52} \sin(2x)$$

Question 6: Consider a simple electrical circuit, where an electromotive force $E = E(t)$ is in series with a resistor of resistance R , and a capacitor of capacitance C , and an inductor of inductance L . The charge $Q = Q(t)$ on the capacitor is a solution of the following ODE

$$L \frac{d^2 Q}{dt^2} + R \frac{dQ}{dt} + \frac{Q}{C} = E.$$

Now consider a specific case where $L = 1\text{H}$, $R = 8\Omega$, $C = 1/15\text{F}$. Assuming $Q(0) = 2$, and assuming that initially there is no current, obtain an expression for $Q(t)$ if $E(t) = 0.1$. Describe the result. Finally, consider the same case, but assume a 'power surge' occurs at $t = 3\text{s}$ such that $E(t) = 0.1 + 8\delta(t - 3)$. Obtain a revised expression for $Q(t)$. Describe the result.

$$\frac{d^2 Q}{dt^2} + 8 \frac{dQ}{dt} + 15Q = 0.1, \quad Q(0) = 2, \quad \frac{dQ}{dt} = 0$$

$$Q = Q_h + Q_p,$$

For Q_h ,

$$\alpha^2 + 8\alpha + 15 = 0 \quad \therefore b^2 - 4ac = 4 > 0 \quad \therefore \alpha = -3 \text{ or } -5$$

$$\therefore Q_h = Ae^{-3t} + Be^{-5t} \quad \therefore \frac{dQ_h}{dt} = -3Ae^{-3t} - 5Be^{-5t}$$

$$\therefore Q(0) = 2 \quad \therefore A + B = 2$$

$$\therefore \frac{dQ}{dt}(0) = 0 \quad \therefore -3A - 5B = 0$$

$$\therefore A = 5, B = -3 \quad \therefore Q_h = 5e^{-3t} - 3e^{-5t}$$

Question 6: Consider a simple electrical circuit, where an electromotive force $E = E(t)$ is in series with a resistor of resistance R , and a capacitor of capacitance C , and an inductor of inductance L . The charge $Q = Q(t)$ on the capacitor is a solution of the following ODE

$$L \frac{d^2 Q}{dt^2} + R \frac{dQ}{dt} + \frac{Q}{C} = E.$$

Now consider a specific case where $L = 1\text{H}$, $R = 8\Omega$, $C = 1/15\text{F}$. Assuming $Q(0) = 2$, and assuming that initially there is no current, obtain an expression for $Q(t)$ if $E(t) = 0.1$. Describe the result. Finally, consider the same case, but assume a 'power surge' occurs at $t = 3\text{s}$ such that $E(t) = 0.1 + 8\delta(t - 3)$. Obtain a revised expression for $Q(t)$. Describe the result.

$$\text{For } Q_p, \text{ let } Q_p = A \quad \therefore Q_p' = 0 \quad \therefore Q_p'' = 0$$

$$\therefore 15A = 0.1 \quad \therefore A = \frac{1}{150}$$

$$\therefore Q = Q_h + Q_p = Q_h + A$$

$$= 5e^{-3t} - 3e^{-1t} + \frac{1}{150}$$

$$\frac{1445}{300}e^{-3t} - \frac{217}{300}e^{-t}$$

$$\frac{d^2 Q}{dt^2} + 8 \frac{dQ}{dt} + 15Q = 0.1 + 8\delta(t-3), \quad Q(0) = 2, \quad \frac{dQ}{dt} = 0$$

$$\therefore L\left\{\frac{d^2 Q}{dt^2}\right\} = sL\{Q\} - Q(0) = sL\{Q\} - 2$$

$$L\left\{\frac{d^2 Q}{dt^2}\right\} = sL\{Q\} - sQ(0) - \frac{dQ}{dt}(0) = sL\{Q\} - 2s$$

$$L\{0.1\} = 0.1 \int_0^\infty e^{-st} dt = \frac{0.1}{s}$$

$$L\{8\delta(t-3)\} = 8L\{\delta(t-3)\} = 8e^{-3s}$$

Question 6: Consider a simple electrical circuit, where an electromotive force $E = E(t)$ is in series with a resistor of resistance R , and a capacitor of capacitance C , and an inductor of inductance L . The charge $Q = Q(t)$ on the capacitor is a solution of the following ODE

$$L \frac{d^2 Q}{dt^2} + R \frac{dQ}{dt} + \frac{Q}{C} = E.$$

Now consider a specific case where $L = 1\text{H}$, $R = 8\Omega$, $C = 1/15\text{F}$. Assuming $Q(0) = 2$, and assuming that initially there is no current, obtain an expression for $Q(t)$ if $E(t) = 0.1$. Describe the result. Finally, consider the same case, but assume a 'power surge' occurs at $t = 3\text{s}$ such that $E(t) = 0.1 + 8\delta(t - 3)$. Obtain a revised expression for $Q(t)$. Describe the result.

$$\therefore s^2 L\{Q\} - 2s + 8s L\{Q\} - 16 + 15 L\{Q\} = \frac{0.1}{s} + 8e^{3s}$$

$$\therefore (s^2 + 8s + 15) L\{Q\} = 2s + 16 + \frac{0.1}{s} + 8e^{3s}$$

$$\begin{aligned} \therefore L\{Q\} &= \frac{2s^2 + 16s + 0.1}{s(s+3)(s+5)} + \frac{8e^{3s}}{(s+3)(s+5)} \\ &= \frac{\frac{1}{150}}{s} + \frac{\frac{244}{60}}{s+3} + \frac{-\frac{244}{100}}{s+5} + \frac{4e^{3s}}{s+3} + \frac{-4e^{3s}}{s+5} \end{aligned}$$

$$\begin{aligned} \therefore Q &= \frac{1}{150} L^{-1}\left\{\frac{1}{s}\right\} + \frac{244}{60} L^{-1}\left\{\frac{1}{s+3}\right\} - \frac{244}{100} L^{-1}\left\{\frac{1}{s+5}\right\} \\ &\quad + 4 L^{-1}\left\{\frac{e^{3s}}{s+3}\right\} - 4 L^{-1}\left\{\frac{e^{3s}}{s+5}\right\} \end{aligned}$$

$$= \frac{1}{150} + \frac{244}{60} e^{-3t} - \frac{244}{100} e^{-5t}$$

$$+ 4 H(t+3) e^{-3t-9} - 4 H(t+5) e^{-3t-15}$$